

4(a)	<table><tr><th>Kingdom</th><th>species may be unicellular</th><th>species may have cell walls</th><th>species may show autotrophic nutrition</th></tr><tr><td>Animalia</td><td>no</td><td>no</td><td>no</td></tr><tr><td>Fungi</td><td>yes</td><td>yes</td><td>no</td></tr><tr><td>Plantae</td><td>no</td><td>yes</td><td>yes</td></tr><tr><td>Protocista</td><td>yes</td><td>yes</td><td>yes</td></tr></table> one mark per correct row	Kingdom	species may be unicellular	species may have cell walls	species may show autotrophic nutrition	Animalia	no	no	no	Fungi	yes	yes	no	Plantae	no	yes	yes	Protocista	yes	yes	yes	4
Kingdom	species may be unicellular	species may have cell walls	species may show autotrophic nutrition																			
Animalia	no	no	no																			
Fungi	yes	yes	no																			
Plantae	no	yes	yes																			
Protocista	yes	yes	yes																			
4(b)	A = metaphase I ; B = telophase I ; A anaphase I C = anaphase II ; A telophase II	3																				

Question	Answer	Marks																														
4(c)	1. Bacteria and Archaea ; plus any two from: <table><tr><td></td><td></td><td>Bacteria</td><td>Archaea</td><td></td></tr><tr><td>2</td><td>cell wall</td><td>peptidoglycan</td><td>no peptidoglycan</td><td>;</td></tr><tr><td>3</td><td>cell membrane</td><td>ester-linked lipids / unbranched hydrocarbon chains</td><td>ether-linked lipids / branched hydrocarbon chains</td><td>;</td></tr><tr><td>4</td><td>spores</td><td>form spores</td><td>do not form spores</td><td>;</td></tr><tr><td>5</td><td>DNA associated proteins</td><td>no histones</td><td>have histones</td><td>;</td></tr><tr><td>6</td><td>ribosomal RNA / rRNA (base sequences) or ribosomes (structure)</td><td colspan="2">are different</td><td>;</td></tr></table>			Bacteria	Archaea		2	cell wall	peptidoglycan	no peptidoglycan	;	3	cell membrane	ester-linked lipids / unbranched hydrocarbon chains	ether-linked lipids / branched hydrocarbon chains	;	4	spores	form spores	do not form spores	;	5	DNA associated proteins	no histones	have histones	;	6	ribosomal RNA / rRNA (base sequences) or ribosomes (structure)	are different		;	3
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4(d)	any two from: 1 RNA or DNA ; 2 single-stranded or double-stranded (nucleic acid) ; 3 AVP ; e.g. how mRNA is produced / enveloped or non-enveloped / linear or circular / which diseases they cause / capsid shape	2																														